

Digital Electronics Lab Report

Submitted by : Neelesh Dorairaj

Roll number : 24EE10015

Lab code : EE29210

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Experiment 6: implementation of a (4×4) -bit multiplier using registers and a down-counter.

Objectives:

1. Use principles of repeated addition to implement a 4 x 4 multiplier.

Apparatus :

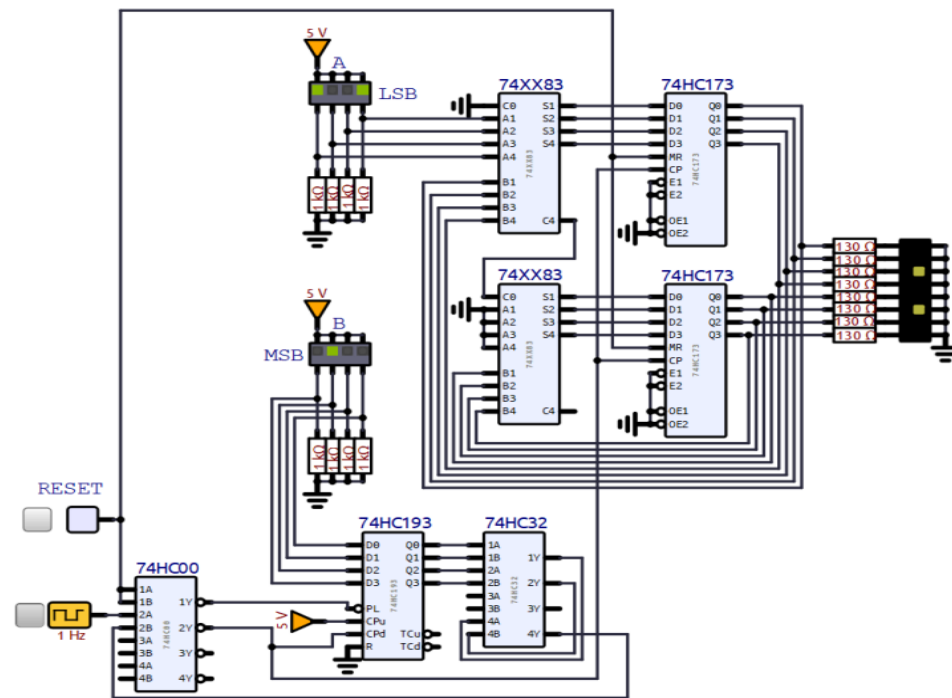
- IC-7483 – 4-bit adder (2)
- IC-74193 – Up/Down counter (1)
- IC-7432 – Quad OR gate (1)
- IC-74374 – 8-bit register (1)
- Resistors – 1 k Ω , (8)

- Breadboard
- DIP Switch – With 4 or more switches (2)
- Connecting wires
- DC Power Supply – 5 V (1)
- Function Generator (1)

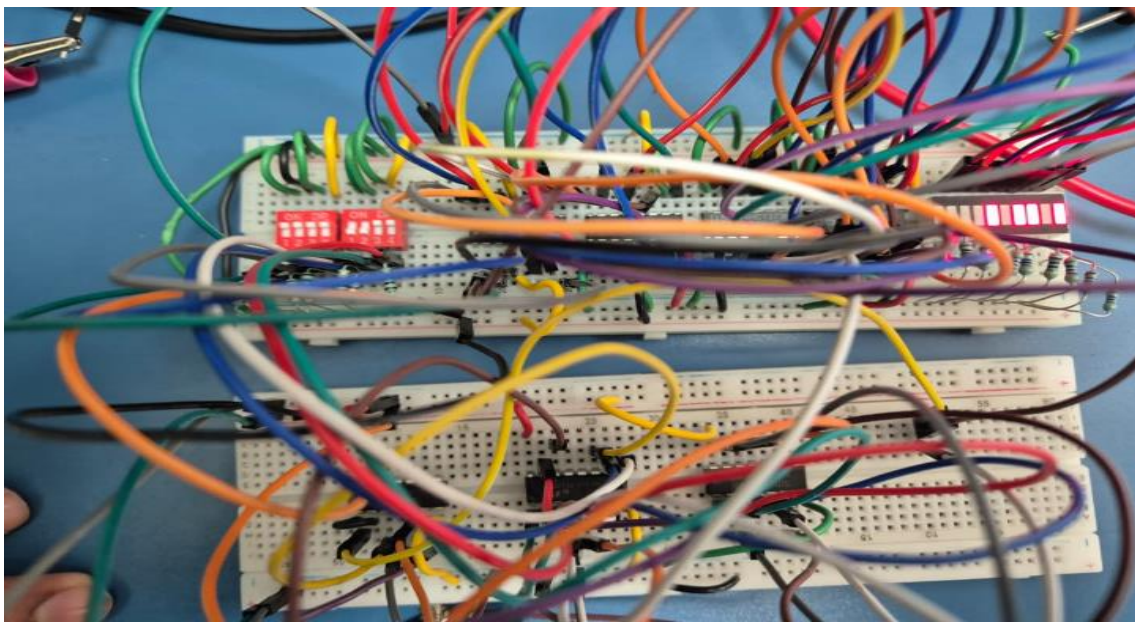
Theory :

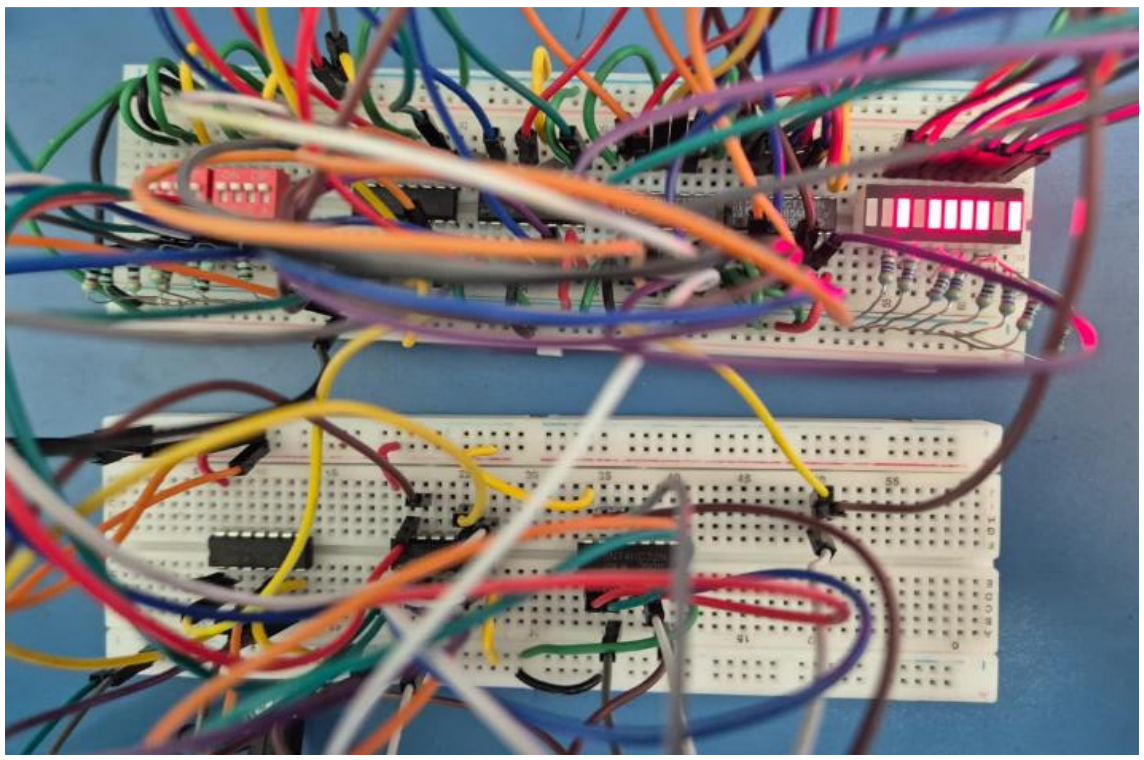
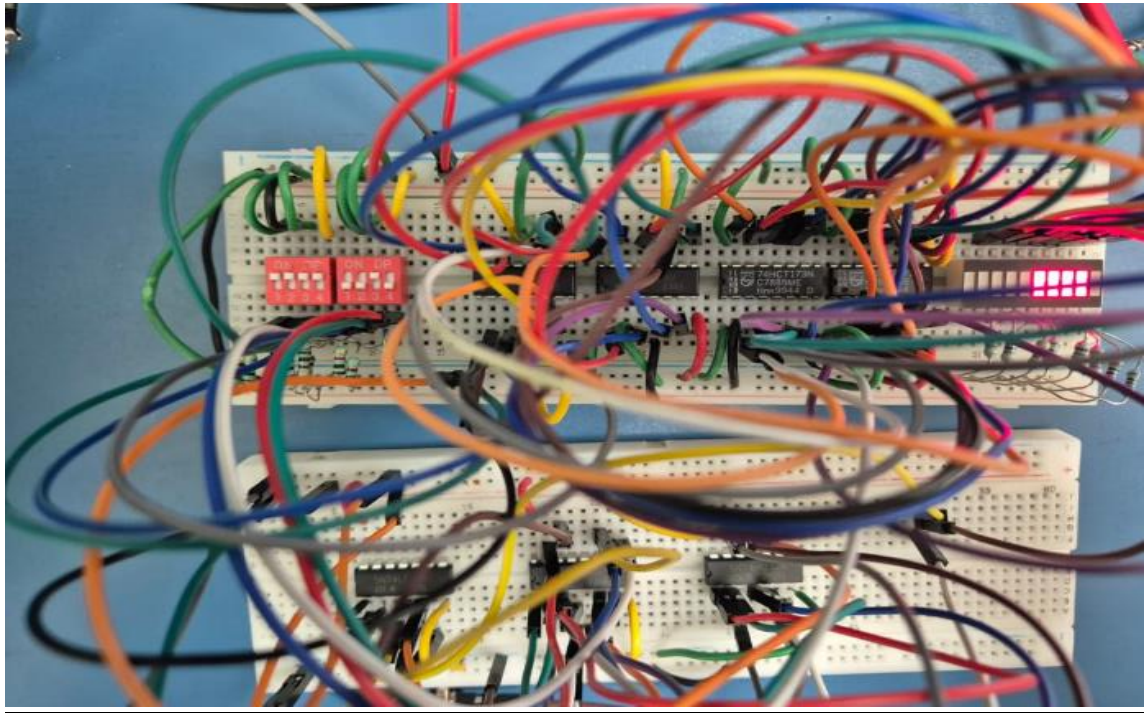
This experiment involves the implementation of a 4×4 -bit binary multiplier using registers and counters based on the principle of repeated addition. In digital systems, multiplication can be performed by repeatedly adding the multiplicand to an accumulator a number of times equal to the multiplier. In this experiment, two 4-bit numbers are multiplied to produce an 8-bit product. The circuit uses IC 7483 4-bit binary adders (two of them combined to form an 8-bit adder), IC 74193 up/down counter, IC 74374 8-bit register, and IC 7432 OR gates to implement the multiplication process. On every clock pulse provided by the function generator, the multiplicand is added to the current value in the product register using the adder circuit. This experiment demonstrates how multiplication operations can be implemented in digital hardware using sequential logic and illustrates the basic principles used in arithmetic units of digital processors.

Circuit Diagram :



Observation :





Conclusion :

In this experiment, a 4×4 -bit binary multiplier was successfully implemented using digital logic components such as IC 7483 adders, IC 74193 down counter, IC 74374 register, and logic gates. The multiplication operation was performed using the principle of repeated addition, where the multiplicand was added repeatedly according to the value stored in the multiplier counter. The down counter controlled the number of additions, while the register stored intermediate and final results. With every clock pulse, the multiplicand was added to the product register until the counter reached zero, producing the final 8-bit product. This experiment demonstrated how arithmetic operations like multiplication can be implemented using sequential digital circuits and helped in understanding the working of adders, registers, counters, and control logic used in digital processors and arithmetic units.